

ASSIGNMENT 4

QUESTION 1

Write a Python program to scrape all available books from the website (<https://books.toscrape.com/>) Books to Scrape – a live site built for practicing scraping (safe, legal, no anti-bot). For each book, extract the following details:

1. Title
2. Price
3. Availability (in stock/out of stock)
4. Star rating (One, Two, Three, Four, Five)

Store the scraped results into a Pandas DataFrame and export them to a CSV file named `books.csv`.

SOLUTION

```
import requests
from bs4 import BeautifulSoup
import pandas as pd

books = []
for i in range(1, 51):
    URL = f"https://books.toscrape.com/catalogue/page-{str(i)}.html"

    response = requests.get(URL)

    html = response.text
    soup = BeautifulSoup(html, "html.parser")

    book_titles = soup.find_all("h3")
    price = soup.find_all("p", class_="price_color")
    availability = soup.find_all("p", class_="instock availability")

    for title, price, availability in zip(book_titles, price, availability):
        book = {
            "title": title.find("a")["title"],
            "price": price.text.strip(),
            "availability": availability.text.strip()
        }
        books.append(book)
    print(f"Page {i} scraped successfully.")

df = pd.DataFrame(books)
df.to_csv("books.csv", index=False)
print(df)
```

OUTPUT

```
Page 1 scraped successfully.
Page 2 scraped successfully.
Page 3 scraped successfully.
Page 4 scraped successfully.
Page 5 scraped successfully.
Page 6 scraped successfully.
Page 7 scraped successfully.
Page 8 scraped successfully.
Page 9 scraped successfully.
```

Page 10 scraped successfully.
 Page 11 scraped successfully.
 Page 12 scraped successfully.
 Page 13 scraped successfully.
 Page 14 scraped successfully.
 Page 15 scraped successfully.
 Page 16 scraped successfully.
 Page 17 scraped successfully.
 Page 18 scraped successfully.
 Page 19 scraped successfully.
 Page 20 scraped successfully.
 Page 21 scraped successfully.
 Page 22 scraped successfully.
 Page 23 scraped successfully.
 Page 24 scraped successfully.
 Page 25 scraped successfully.
 Page 26 scraped successfully.
 Page 27 scraped successfully.
 Page 28 scraped successfully.
 Page 29 scraped successfully.
 Page 30 scraped successfully.
 Page 31 scraped successfully.
 Page 32 scraped successfully.
 Page 33 scraped successfully.
 Page 34 scraped successfully.
 Page 35 scraped successfully.
 Page 36 scraped successfully.
 Page 37 scraped successfully.
 Page 38 scraped successfully.
 Page 39 scraped successfully.
 Page 40 scraped successfully.
 Page 41 scraped successfully.
 Page 42 scraped successfully.
 Page 43 scraped successfully.
 Page 44 scraped successfully.
 Page 45 scraped successfully.
 Page 46 scraped successfully.
 Page 47 scraped successfully.
 Page 48 scraped successfully.
 Page 49 scraped successfully.
 Page 50 scraped successfully.

	title	price	availability
0	A Light in the Attic	£51.77	In stock
1	Tipping the Velvet	£53.74	In stock
2	Soumission	£50.10	In stock
3	Sharp Objects	£47.82	In stock
4	Sapiens: A Brief History of Humankind	£54.23	In stock
..	...	...	...
995	Alice in Wonderland (Alice's Adventures in Won...	£55.53	In stock
996	Ajin: Demi-Human, Volume 1 (Ajin: Demi-Human #1)	£57.06	In stock
997	A Spy's Devotion (The Regency Spies of London #1)	£16.97	In stock
998	1st to Die (Women's Murder Club #1)	£53.98	In stock
999	1,000 Places to See Before You Die	£26.08	In stock

[1000 rows x 3 columns]

QUESTION 2

Write a Python program to scrape the IMDB Top 250 Movies list (<https://www.imdb.com/chart/top/>). For each movie, extract the following details:

1. Rank (1-250)
2. Movie Title
3. Year of Release
4. IMDB Rating

SOLUTION

```
from selenium import webdriver
from selenium.webdriver.common.by import By
import pandas as pd

driver = webdriver.Firefox()
driver.get("https://www.imdb.com/chart/top/")

# ul.ipc-metadata-list
movies = driver.find_elements(By.CSS_SELECTOR, "ul.ipc-metadata-list li")
rows = []
for movie in movies:
    obj = {}
    title_selector = ".ipc-title__text.ipc-title__text--reduced"
    title = movie.find_element(By.CSS_SELECTOR, title_selector).text

    rank = title.split(".")[0]
    obj["Rank"] = rank

    year_selector = ".cli-title-metadata-item"
    year = movie.find_element(By.CSS_SELECTOR, year_selector).text

    rating_selector = ".ipc-rating-star--rating"
    rating = movie.find_element(By.CSS_SELECTOR, rating_selector).text

    obj["Rank"] = rank
    obj["Title"] = title.split(". ")[1]
    obj["Year"] = year
    obj["Rating"] = rating
    rows.append(obj)

data = pd.DataFrame(rows)
print(data)

data.to_csv("imdb_top_movies.csv", index=False)
```

OUTPUT

	Rank	Title	Year	Rating
0	1	The Shawshank Redemption	1994	9.3
1	2	The Godfather	1972	9.2
2	3	The Dark Knight	2008	9.1
3	4	The Godfather Part II	1974	9.0
4	5	12 Angry Men	1957	9.0
..	...	...	...	...
245	246	The Help	2011	8.1
246	247	Groundhog Day	1993	8.0
247	248	Demon Slayer: Kimetsu no Yaiba - The Movie: Mu...	2020	8.2
248	249	Drishyam	2015	8.2

[250 rows x 4 columns]

QUESTION 3

Write a Python program to scrape the weather information for top world cities from the given website (<https://www.timeanddate.com/weather/>). For each city, extract the following details:

1. City Name
2. Temperature
3. Weather Condition (e.g., Clear, Cloudy, Rainy, etc.)

Store the results in a Pandas DataFrame and export it to a CSV file named `weather.csv`.

SOLUTION

```
import requests
from bs4 import BeautifulSoup
import pandas as pd

URL = "https://www.timeanddate.com/weather/"

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/58.0.3029.110 Safari/537.3"
}

response = requests.get(URL, headers=headers)
soup = BeautifulSoup(response.content, "html.parser")

td = soup.find_all("td")

rows = []
row = {}

for idx, i in enumerate(td):
    if idx % 4 == 0:
        row = {}
        row["City"] = i.text.strip()
        if i.text.strip().endswith("*"):
            row["City"] = row["City"][:-1].strip()
    elif idx % 4 == 1:
        try:
            row["Local Time"] = i.text.split()[1].strip()
        except:
            row["Local Time"] = None
    elif idx % 4 == 2:
        try:
            img = i.find("img")
            row["Weather"] = img["alt"] if img else None
        except:
            row["Weather"] = None
    elif idx % 4 == 3:
        row["Temperature"] = i.text.strip()
        # completed row → append to rows
        rows.append(row)

data = pd.DataFrame(rows)
print(data)
```

```
data.to_csv("weather.csv", index=False)
```

```
# 0: Accra  
# 1: सोम 08.58  
# 2:  
#    img: Overcast. Mild.  
# 3: 24 °C  
#
```

OUTPUT

	City	Local Time	Weather	Temperature
0	Accra	09.09	Overcast. Mild.	24 °C
1	Kuala Lumpur	17.09	Passing clouds. Hot.	33 °C
2	Addis Ababa	12.09	Fog. Cool.	15 °C
3	Kuwait City	12.09	Sunny. Extremely hot.	41 °C
4	Adelaide	18.39	Cool.	16 °C
..	...	...	...	...
135	Zagreb	11.09	Sunny. Mild.	20 °C
136	Kiritimati	23.09	Passing clouds. Warm.	27 °C
137	Zürich	11.09	Cool.	16 °C
138	Kolkata	14.39	Scattered clouds. Warm.	31 °C
139		None	None	

[140 rows x 4 columns]